INCH-POUND

MIL-B-43021E
11 June 1992
SUPERSEDING
MIL-B-43021D
31 May 1978

MILITARY SPECIFICATION

BINDER, LOAD, CARGO TIE-DOWN, 10,000 POUND CAPACITY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type and size load binder of 10,000 pound capacity.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be used in improving this document should be addressed to: U.S. Army Natick, Research, Development, and Engineering Center, Natick, MA 01760-5017 by using the Standardization Documents Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1670

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SPECIFICATIONS

FEDERAL

PPP-B-585 - Boxes, Wood, Wirebound
PPP-B-601 - Boxes, Wood, Cleated-Plywood
PPP-B-636 - Boxes, Shipping, Fiberboard

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-130 - Identification Marking of U.S. Military Property
MIL-STD-147 - Palletized Unit Loads

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

U.S. ARMY NATICK RESEARCH, DEVELOPMENT, AND ENGINEERING CENTER

11-1-901 - Binder, Load, Cargo Tie-Down, 10,000 lb Capacity; Assembly

(Copies of drawings are available from the U.S. Army Natick Research, Development, and Engineering Center, ATTN: STRNC-UX, Natick, MA 01760-5017.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the issue of the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3951 - Standard Practice for Commercial Packaging

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(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.2 Materials and components. Materials and components shall be as specified herein and on the applicable drawing. It is encouraged that recycled material be used when practical as long as it meets the requirements of this specification.

3.3 Design and construction. Materials, components, protective finish, marking, construction and assembly shall conform to Drawing 11-1-901.

3.4 Performance.

3.4.1 Limit load. The assembled load binder and components when disassembled shall show no visible evidence of release, fracture, weld separation, or permanent deformation when tested to a load of 10,000 pounds as specified in 4.4.5.

3.4.2 Ultimate load. The assembled load binder shall withstand an ultimate load of 13,000 pounds without a release or failure when tested as specified in 4.4.5. Deformation, but not rupture of parts shall be allowed.

3.5 Identification marking. Each load binder shall be marked for identification with the capacity, 10,000 LB, manufacturer's name or readily identifiable trade mark, and the Part Number 11-1-901. Part numbering shall be in accordance with MIL-STD-130. The markings shall be permanently indented, located as shown on the drawing, and of such size as to be clearly legible.

3.6 Workmanship. All components and assemblies of the load binder assembly shall be free from dirt and other extraneous material, burrs, slivers, dents, cracks, fractures and damaged welds. External surfaces shall

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be free from sharp edges and corners. Components shall not be buckled, loose, dented, missing, malformed or misaligned. The load binders shall conform to the quality of product established by this specification and the occurrence of defects shall not exceed the applicable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Responsibility for dimensional requirements. Unless otherwise specified in the contract or purchase order, the contractor is responsible for ensuring that all specified dimensions have been met. When dimensions cannot be examined on the end item, inspection shall be made at any point, or at all points in the manufacturing process necessary to ensure compliance with all dimensional requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 3.1 and 6.2), it shall be examined for the defects specified in 4.4.3 and 4.4.4 and shall be tested for the characteristics specified in 4.4.5. Any nonconformance shall be cause for rejection of the first article.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

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4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.4.2 In-process inspection. Inspection of subassemblies shall be made to ascertain that construction details which cannot be examined in the finished product are in accordance with specified requirements. The Government reserves the right to exclude from consideration for acceptance, any material or service for which in-process inspection has indicated nonconformance.

4.4.3 End item visual examination. The end items shall be examined for the defects listed in table I. The lot size shall be expressed in units of load binders. The sample unit shall be one load binder. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5 for major defects and 6.5 for total (major and minor combined) defects.

TABLE I. End item visual defects

Examine	Defect	Classification Major	Minor
Finish, protective	Any breaks through coating of metal		201
	Any area of rust or corrosion	101	
	Dirt, oil, grease, or other foreign matter		202
Construction and workmanship	Any component missing	102	
	Any component cracked, fractured or otherwise damaged, including damaged welds	103	
	Any sharp edge, burr or metal sliver	104	
	Any part not assembled or joined as specified	105	
Identification marking	Missing, incomplete not legible, incorrect or not located or applied as specified		203

4.4.4 End item dimensional examination. The end items shall be examined for conformance to the dimensions specified on Drawing 11-1-901. Only those dimensions that can be evaluated without damaging or disassembling the end items shall be examined. Any dimension not within the specified tolerance shall be classified as a defect. The lot size shall be expressed in units of

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load binders. The sample unit shall be one load binder. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

4.4.5 End item testing. The end items shall be tested as specified in 4.5.1 and 4.5.2 for compliance with 3.4.1 and 3.4.2. The lot size shall be expressed in units of load binders. The sample unit shall be one completely fabricated load binder. The inspection level for the limit load test shall be S-2. The results shall be reported as pass or fail. Any failure during the limit load test shall be cause for rejection of the lot. Upon completion of the limit load test, two of the same samples previously subjected to the limit load test shall be retested for ultimate load and the results reported as pass or fail (if during the ultimate load test, failure of a sample unit occurs under 13,000 pounds. The testing shall then be performed on all of the samples submitted for the limit load test). In this retest, if the average ultimate load for all samples falls below 12,000 pounds or if a single sample fails below 10,000 pounds, the lot shall be rejected.

4.4.6 Packaging examination. The fully packaged end items shall be examined for the defects listed below. The lot size shall be expressed in units of shipping containers. The sample unit shall be one shipping container fully packaged. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application
Materials	Any component missing, damaged, or not as specified
Workmanship	Inadequate application of components, such as: incomplete sealing or closure of flap, improper taping, loose strapping, or inadequate stapling Bulged or distorted container
Content	Number per container is more or less than required

4.4.7 Palletization examination. The fully packaged and palletized end items shall be examined for the defects listed below. The lot size shall be expressed in units of palletized unit loads. The sample unit shall be one palletized unit load, fully packaged. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

Examine	Defect
Finished dimensions	Length, width, or height exceeds specified maximum requirement

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Palletization	Pallet pattern not as specified Load not bonded as specified
Weight	Exceeds maximum load limits
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application

4.5 Methods of inspection.

4.5.1 Limit load test. The binder shall be secured in a suitable tensile testing device and subjected to a static load of 10,000 pounds applied for not less than 30 seconds. Direction of load shall be as shown on Drawing 11-1-901. After load relaxation, the hooks shall be disassembled from the binder and all components examined for evidence of fracture, weld separation and permanent deformation. Any deformation in excess of 1/32 inch from the drawing dimensions, in any dimension parallel to the direction of stress, shall be classified as visible permanent deformation.

4.5.2 Ultimate load test. The binder shall be secured in a suitable tensile testing device and subjected to a load of 13,000 pounds applied gradually in the direction shown on Drawing 11-1-901. The load shall be removed immediately upon reaching 13,000 pounds. After load relaxation, the binder shall be examined for evidence of damage as specified in 3.4.2. Any failure at or above 13,000 pounds tensile load shall not constitute a failure. The ultimate load test is considered to be a destructive test. Units subjected to the ultimate load test or any parts thereof shall not be recycled or delivered as part of the quantity of units on order.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or commercial, as specified (see 6.2).

5.1.1 Level A preservation. Each binder shall be unit packed in a fiberboard box conforming to style OPF or RSC, grade W6c or grade 200 of PPP-B-636. Inside dimensions of each box shall approximate 13-1/4 inches in length, 3-5/8 inches in width, and 1-1/8 inches in depth. Approximate dimensions are furnished as a guide only. Each interior box shall be closed method IV and reinforced method III in accordance with the appendix of PPP-B-636.

5.1.2 Commercial preservation. Each binder shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

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5.2.1 Level A packing. Seventy-two binders, preserved as specified in 5.1, shall be packed in a shipping container conforming to overseas type, style A or J, type 2 load of PPP-B-601; or class 3, style optional, type 2 load of PPP-B-585. Level A unit packs shall be packed on end, 6 in length, 12 in width, and 1 in depth within a shipping container. Inside dimensions of each shipping container shall approximate 23-1/4 inches in length, 16-1/2 inches in width, and 13-1/2 inches in depth. Approximate dimensions are furnished as a guide only. Each shipping container shall be closed and reinforced in accordance with the appendix of the applicable container specification.

5.2.2 Level B packing. Seventy-two binders, preserved as specified in 5.1, shall be packed in a shipping container conforming to domestic type, style A or J, type 2 load of PPP-B-601; or class 1, style optional, type 2 load of PPP-B-585. Level A unit packs shall be packed on end, 6 in length, 12 in width, and 1 in depth within a shipping container. Inside dimensions of each shipping container shall approximate 23-1/4 inches in length, 16-1/2 inches in width, and 13-1/2 inches in depth. Approximate dimensions are furnished as a guide only. Each shipping container shall be closed and reinforced in accordance with the appendix of the applicable container specification.

5.2.3 Commercial packing. Binders preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

5.3 Palletization. When specified (see 6.2), binders, packed as specified in 5.2, shall be palletized in accordance with load type I of MIL-STD-147. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means C and D. Pallet patterns shall be in accordance with the appendix of MIL-STD-147.

5.4 Marking. In addition to any special marking required by the contract or purchase order, unit packs, shipping containers, and palletized unit loads shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The load binder is designed for use as a tie-down tensioning device in rigging general cargo and equipment for transport and air delivery.

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6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. When a first article is required (see 3.1, 4.3, and 6.3).
d. Levels of preservation and packing (see 5.1 and 5.2).
e. When palletization is required (see 5.3).

6.3 First article. When a first article sample is required, it shall be inspected and approved under the appropriate provisions of Federal Acquisition Regulation (FAR) 52.209-4. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.4 Subject term (key word) listing.

Air delivery
Rigging
Tensioning device

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:

Army - GL
Air Force - 99

Review activities:

Army - TR
Air Force - 11

User activity:

Navy - MC

Preparing activity:

Army - GL

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-B-43021E	1992 June 11

3. DOCUMENT TITLE
BINDER, LOAD, CARGO TIE-DOWN, 10,000 POUND CAPACITY

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON/DSN
U.S. Army Natick RD&E Center	508-651-4531	256-4531
c. ADDRESS (Include Zip Code) Commander, U.S. Army Natick RD&E Center ATTN: STRNC-UXT Natick, MA 01760-5017	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

DD Form 1426, OCT 89 NATICK OP3, JUL 91 Previous editions are obsolete 198/290

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